0.1 Quadratic Integer Rings

Definition 0.1.1. Let D be a rational number which is not perfect square in $\mathbb{Q}$. Define a *Quadratic Field*:

$$\mathbb{Q}(\sqrt{D}) \overset{\text{def}}{=} \{a + b\sqrt{D} \mid a, b \in \mathbb{Q}\}$$

It is *field* with additive identity 0 and multiplicative identity 1, and the additive inverse is

$$-(a + b\sqrt{D}) = -a - b\sqrt{D}$$

and the multiplicative inverse is

$$(a + b\sqrt{D})^{-1} = \frac{a - b\sqrt{D}}{a^2 - Db^2}$$

And other conditions will be shown just calculation. Moreover, it is subfield of $\mathbb{C}$.

Definition 0.1.2. Let D be a square-free integer. That is, for any prime p, p^2 cannot divide D.
And, put

$$\omega \overset{\text{def}}{=} \begin{cases} \sqrt{D} & D \equiv 2, 3 \bmod 4 \\ \dfrac{1+\sqrt{D}}{2} & D \equiv 1 \bmod 4 \end{cases}, \quad \overline{\omega} \overset{\text{def}}{=} \begin{cases} -\sqrt{D} & D \equiv 2, 3 \bmod 4 \\ \dfrac{1-\sqrt{D}}{2} & D \equiv 1 \bmod 4 \end{cases}$$

Define a *Quadratic Integer Ring* as:

$$\mathbb{Z}[\omega] \overset{\text{def}}{=} \{a + b\omega \mid a, b \in \mathbb{Z}\}$$

It is a *Ring*. In case of $D \equiv 2, 3 \bmod 4$, it satisfies the conditions of ring clearly. In case of $D \equiv 1 \bmod 4$, given $a + b\omega, c + d\omega \in \mathbb{Z}[\omega]$, observe that:

$$\begin{aligned} \left(a + b\frac{1+\sqrt{D}}{2}\right) \cdot \left(c + d\frac{1+\sqrt{D}}{2}\right) &= ac + (ad + bc)\left(\frac{1+\sqrt{D}}{2}\right) + bd\left(\frac{1+D+2\sqrt{D}}{4}\right) \\ &= ac + bd \cdot \frac{D-1}{4} + (ad + bc + bd)\left(\frac{1+\sqrt{D}}{2}\right) \in \mathbb{Z}[\omega] \end{aligned}$$

so, it is closed under the multiplication. And, the other conditions are also satisfied, just simple calculation.
And, more observation: $\overline{\omega}$ is the other root of minimal polynomial of ω. Specifically,

$$x^2 - D \in \mathbb{Z}[x]$$

is the minimal polynomial of $\sqrt{D}$, having roots as $\sqrt{D}$ and $-\sqrt{D}$. And,

$$(2x-1)^2 - D = 4x^2 - 4x + 1 - D \in \mathbb{Z}[x]$$

is the minimal polynomial of $\dfrac{1+\sqrt{D}}{2}$, having roots as $\dfrac{1+\sqrt{D}}{2}$ and $\dfrac{1-\sqrt{D}}{2}$.

Theorem 0.1.3. The field Norm $N : \mathbb{Q}(\sqrt{D}) \to \mathbb{Q}$ defined by $N(a + b\sqrt{D}) = a^2 - Db^2$ preserves multiplication.

Proof. Let $a + b\sqrt{D}, c + d\sqrt{D} \in \mathbb{Q}(\sqrt{D})$ be given. Then,

$$\begin{aligned} N\big((a + b\sqrt{D}) \cdot (c + d\sqrt{D})\big) &= N\big(ac + bdD + (ad + bc)\sqrt{D}\big) \\ &= (ac + bdD)^2 - D(ad + bc)^2 \\ &= a^2c^2 + b^2d^2D^2 + 2abcdD - D(a^2d^2 + b^2c^2 + 2abcd) \\ &= a^2c^2 + b^2d^2D^2 - D(a^2d^2 + b^2c^2) \\ &= (a^2 - Db^2)(c^2 - Dd^2) \\ &= N(a + b\sqrt{D}) \cdot N(c + d\sqrt{D}). \end{aligned}$$

□

Theorem 0.1.4. In $\mathbb{Z}[\omega]$, an element α is a unit if and only if $N(\alpha) = \pm 1$.